

URL SHORTENER

Functional requirements

1. Account creation -Register/Login
2. Convert long URL to short URL
3. History of converted URLs
4. Redirect to the original URL
5. Custom URL creation (optional)

Non -Functional requirements

1. Unique shortened URL
2. Availability
3. Responsive
4. Low latency
5. Consistency

Estimations

- 10 million daily users
- Each user roughly shortens 2 URLs daily
- URLs shortened daily - 20 million
- 20 million writes per day => **2 billion redirects per day**
- Metadata -> created timestamp, expiration, user ID
=> **300 bytes (approx)**
- Total storage occupied for one day - 20 million * 300
=> 6 GB per day
- Total storage required for 1 year - 6 * 365
=> **2.2 TB per year**
- 20 million hot URLs. Original URL - 300 bytes
cache size required = 20M * 300
=> 6 GB
round off to 8-10 GB

Architecture

